

THE NEXT Y2K

Yes, the techies have done it again—brought the world to the brink of another disaster, because they failed to foresee the exponential growth of the Internet! Read on to learn about why IPv4 shows threatening signs of becoming the next Y2K disaster.

The Internet has become an essential part of our lives. Everybody is on the Net, surfing with different types of devices: PCs, laptops, iPads, mobile phones, etc. Each of these needs a unique IP address on the Internet, when directly connected. The present IP (Internet Protocol) addressing system, IPv4, uses a 32-bit number, represented in 4 octets—for example, 192.16.1.2.

The problem

Since IPv4 addresses are 32-bit numbers, there are only 2^{32} , or just over four billion, possible addresses available—and they have all been allocated. On 3rd February 2011, the Internet Assigned Numbers Authority (IANA), which controls the world-wide distribution of IP addresses, allocated the final pool of IPv4 address space to the five Regional Internet Registries (RIRs). This means that there are no longer any IPv4 addresses available for allocation from the IANA to the five RIRs.

So does this mean the end of the Net? Well, not really. Luckily, this problem was foreseen, and an alternative to IPv4 developed. The new addressing scheme, IPv6, uses 128-bit numbers for addresses—hence, there are 2^{128} individual addresses available (approximately 3.4×10^{38}). All new IP addresses from now will be IPv6.

What, then, is the 'next Y2K'? The problem is that IPv4 is NOT directly compatible with IPv6. This means that devices connected via IPv4 (which make up most of the existing Internet) cannot communicate directly with systems connected using IPv6 (and vice versa). If your website is on an IPv4 connection, it won't get traffic from IPv6 sites/users and the reverse is also true. Getting traffic is essential to any business on the Net; nobody can afford to lose IPv6 users. Hence, to continue offering customers the same level of service, every organisation needs to be planning IPv6 deployment now.

The IPv6 compliance scenario

All major US government federal agencies have migrated their infrastructure (network backbones) to using IPv6, and all agency networks were enabled to interface with this infrastructure with

effect from June 2008. The US government's Chief Information Officers require that all new IT procurements be IPv6 compliant. An IPv6-compliant product or system must be able to receive, process, and transmit or forward (as appropriate) IPv6 packets, and should interoperate with other systems and protocols in both IPv4 and IPv6 modes of operation.

The US Defence Department, with its \$30 billion budget, has been buying only IPv6-compliant networking gear since October 2003, and had already achieved full IPv6 compliance in 2008. ICANN, the company that handles the domain names and IP addresses for the world, has also enabled its servers to be IPv6-compliant. There has been significant IPv6 adoption in Japan and Korea, with production networks and consumers paying for IPv6-based services. China is spending millions developing a new backbone network that is reportedly going to be IPv6. The European Union (EU) has also heavily financed R&D in IPv6 backbone networks and innovative services that leverage many of the beneficial features of IPv6.

On 8th June 2011, Google, Facebook, Yahoo!, Akamai and Limelight Networks are among some major organisations that will offer their content over IPv6 for a 24-hour 'test flight'. The goal of this Test Flight Day is to motivate organisations across the industry—Internet Service Providers, hardware manufacturers, operating system vendors and Web companies—to prepare their services for IPv6, to ensure a successful transition as IPv4 addresses run out.

India will start using IPv6 (Internet Protocol version 6) from March 2012, according to a new roadmap released by the Indian government. All telecom companies and ISPs will have to be IPv6-compliant by the end of next year, and offer IPv6 services thereafter. Federal and state government ministries, departments and public sector companies will switch over to IPv6 services by March 2012.

So what do I do?

It is imperative for any organisation to be IPv6-ready, in order to remain competitive. The first step is to assess where your